Unofficial Liiga API (v2) Reference

This document provides a consolidated reference for the undocumented RESTful API utilized by `liiga.fi` after its late-2024 digital overhaul.

1. Global Request Requirements

To successfully query these endpoints and avoid 403 Forbidden errors, blocks, or connection resets, your HTTP requests **must** simulate legitimate browser behavior.

- **Base URL:** `https://liiga.fi/api/v2`
- **Authentication:** No persistent API keys or OAuth tokens are required for the `v2` wrapper API.
- **Mandatory Headers:**
  - `User-Agent` : Must be spoofed as a modern desktop browser (e.g., `Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36...`). *Do not use default library agents like `python-requests` .*
  - `Referer` : `https://liiga.fi/` (Crucial for bypassing CORS policies).
  - `Accept` : `text/html,application/xhtml+xml,application/xml;q=0.9,image/webp,*/*;q=0.8`

2. API Endpoints Table

HTTP Method	Endpoint Path	Parameter Type	Required Parameters & Values	Description	How to Use / Context

GET	`/games`	Query	`?tournament=[type]`
			`&season=[year]`
			
			Valid Tournaments:
			`runkosarja` , `playoffs` , `chl` , `valmistavat_ottelut`
				Returns master schedule and high-level game metadata.	**Index Endpoint:** Use this to programmatically map and extract the unique integer `id` for every game in a season, which is required for subsequent match-specific queries.

			Season: Closing year (e.g., `2025` for 24-25)		
GET	`/schedule`	Query	`?tournament=[type]` `&season=[year]`	Alternative scheduling endpoint with a different JSON structure.	Best used for automated calendar integrations or generating flat CSV exports of the season schedule.
GET	`/games/{season}/{id}`	Path	`{season}` : Year (e.g., 2025) `{id}` : Integer Game ID (e.g., 86)	Returns logistical match metadata.	Query to extract line combinations (*ketjunumerot*), penalty tracking (*rangaistukset*), referee assignments, and exact match start times.
GET	`/games/stats/{season}/{id}`	Path	`{season}` : Year `{id}` : Integer Game ID	Returns advanced match statistics powered by Wisehockey.	Query to analyze single-game team metrics (Expected Goals/xG, Corsi, faceoff percentages) and individual player performance metrics for that specific night.
GET	`/shotmap/{season}/{id}`	Path	`{season}` : Year `{id}` : Integer Game ID	Returns a discrete array of all physical shot events for a game.	Use to build spatial heatmaps. Each object contains exact x/y coordinates, shot outcome, shooter database ID, and the calculated xG probability value.
GET	`/players/stats/summed/{start_year}/{end_year}/{tournament}/true`	Path	`{start_year}` : Year `{end_year}` : Year `{tournament}` : String (e.g., `runkosarja`)	Returns cumulative, aggregated player statistics across a chronological period.	Query for dense, season-long tracking metrics like `time_of_ice` , `powerplay_seconds_total` , and advanced defensive metrics like `player_xga_vs_time_of_ice_per_minute` . *(Note: Requires extracting internal Player IDs from public frontend profile URLs to map to real names).*

3. Special Case: Team Standings (Frontend Wrapper)

While the raw API handles granular game data, league-wide team aggregations are routed through a separate frontend query structure.

HTTP Method	URL Structure	Required Parameters & Values	Description	How to Use / Context

GET	`https://www.liiga.fi/en/stats/teamstats/season`	`?sarja=runkosarja`
		`&tilasto=sarjataulukko`
		`&kaudesta=[start_year]`
		`&kauteen=[end_year]`
			Returns comprehensive league standings and team-level metrics.	Query to get the league table combined with embedded advanced predictive metrics like Expected Goals For (xGF), Expected Goals Against (xGA), Team Skating Distance, and Puck Control percentages (PC%).

4. Handling Responses & Data Anomalies

- **Fault Tolerance Required:** Be prepared for corrupted or malformed data resulting from sensor dropouts or API regressions.
- **JSON Parsing:** Always preemptively clean null byte strings (e.g., replacing `'null'` with `'"placeholder"'`) before mapping to dataframes to avoid fatal exceptions.
- **Legacy Data (Pre-2024):** If testing older data endpoints, beware of JSONP padding strings (e.g., `jQuery172...(...)`) wrapping the payload. These must be stripped via regex before parsing standard JSON.